Masari

HCAI Term Project 2026

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1. The Two AI-Generated Concepts

1.1 What must be included for each tool

We used the same prompt with two different AI tools, Claude and Gemini, to generate alternative concepts for our AI-based guidance system for Grade 9 students transitioning into the Egyptian Baccalaureate system. The two drafts are presented separately and compared using the same criteria to identify their strengths, limitations, and potential contributions to the final design.

1.2 Claude — "Masar: The Transparent Pathway Guide"

Part A — Description of the original output

Field	Content of the original output (descriptive summary)
Tool	Claude - Model/version: **Sonnet 5.5**
Concept name	"Masar: The Transparent Pathway Guide"
Original output	Full text supplied as `Masari_Claude_Week_02.md`, retained as the original concept draft.
Main interaction flow	A detailed, flexible journey covering system introduction, pathway exploration, comparison, interest assessment, profile review, recommendations, explanation inspection, follow-up questions, what-if scenarios, and decision reflection.
Key screens / features	Home screen, system explanation cards, pathway comparison, assessment questions, suggestion cards, a reasoning panel, conversational guidance, and a decision notebook..
Transparency mechanisms	Source labels and verification dates, visible reasons behind recommendations, confidence indicators, editable answers, adjustable importance weights, and explanations of uncertainty.
Error / recovery handling	Unverified policy information is flagged. Skipped answers are not guessed, and the system communicates uncertainty when the available information is insufficient.
Privacy / control	No required personal identification, optional session saving, data deletion controls, and the ability to skip questions or leave the process at any time.

Part B — Strengths & Weaknesses

	Observations
Strengths	• Provides a detailed interaction flow from the welcome screen to the final reflection stage. • Makes the reasoning behind pathway suggestions visible and open to inspection. • Allows students to edit answers, change the importance of different factors, and test alternative scenarios. • Includes a decision notebook to help students organize their thoughts and discuss options with parents or teachers. • Emphasizes source verification, uncertainty, student privacy, and responsible guidance.
Weaknesses	• The large number of interaction steps and screens may increase development effort for a university term project. • The assessment, scoring rules, reasoning panel, conversational interface, and notebook require careful integration. • The proposed recommendation logic and confidence indicators need implementation and testing before their reliability can be assessed. • Official Baccalaureate information must be verified before it can be presented as factual guidance.

1.3 Gemini — “Masari: An Explainable AI Transition Guide for the Egyptian Baccalaureate”

Part A — Description of the original output

Field	Content of the original output (descriptive summary)
Tool	Gemini (Google). Model/version: **Gemini Flash 3.6 Low**
Concept name	“Masari: An Explainable AI Transition Guide for the Egyptian Baccalaureate”
Original output	Full text supplied as Masari_Gemini_Week_02.md, retained as the original concept draft.
Main interaction flow	Five stages: understanding the system, comparing pathways, completing a self-assessment, reviewing explainable recommendations, and exploring alternatives through an interactive guidance interface.
Key screens / features	System overview, side-by-side pathway comparison, interactive assessment, recommendation dashboard, and guidance sandbox with a decision summary.
Transparency mechanisms	Visible recommendation factors, editable assessment inputs, adjustable preference weights, and explanations showing how student responses contribute to matching scores.
Error / recovery handling	Unverified Ministry policies are flagged, and the system communicates limitations instead of presenting unsupported information as certain.
Privacy / control	Avoidance of personally identifiable information, session-based data handling, adjustable preferences, and clear statements about the system’s advisory limitations.

Part B — Strengths & Weaknesses

	Observations
Strengths	• Presents a relatively short, easy-to-follow five-stage interaction flow. • Provides a clear recommendation dashboard with visible factors explaining pathway matches. • Supports user control through editable inputs and adjustable preference weights. • Includes a conversational guidance interface for exploring alternative scenarios. • Defines a proposed technical architecture using React, FastAPI, a deterministic scoring component, and an LLM endpoint.
Weaknesses	• The original design includes percentage-based affinity scores that may imply more precision than the scoring method can justify. • The proposed matching weights and pathway representations require validation before being used for recommendations. • The technical scope includes several connected components that may require significant integration and testing. • The quality of its guidance depends on the accuracy and verification of the official information used. • The proposed evaluation targets are goals for future testing, not demonstrated results.

2. Side-by-Side Comparison

Criterion	Claude concept	Gemini concept	Preliminary observation
Core interaction flow	A detailed, flexible journey covering assessment, recommendations, explanations, and decision reflection.	A compact five-stage flow covering understanding, exploration, assessment, recommendation, and guidance.	Claude provides more interaction detail, while Gemini offers a simpler high-level structure.
Ease of use	Includes multiple screens and controls that may require careful navigation design.	Uses a shorter sequence and a centralized recommendation dashboard.	The final design should preserve a clear sequence without restricting students to a rigid path.
User Control and Agency and source verification	Includes a reasoning panel, visible contributing factors, confidence notes, and an option to inspect the scoring rules.	Includes recommendation factor breakdowns and controls for inspecting and adjusting matching preferences.	Both prioritize explainability. The final system should make the reasons understandable without implying that the recommendations are objectively correct.
Recommendation logic	Proposes rule-based scoring, documented pathway relevance values, and qualitative fit bands.	Proposes a weighted matching formula and percentage-based affinity scores.	A transparent, rule-based approach with carefully communicated uncertainty is preferable for the initial prototype.
Student control	Allows students to edit answers, adjust importance weights, hide factors, and explore what-if scenarios.	Allows students to modify assessment inputs and preference weights and explore alternatives.	Both support user agency. These controls should be retained where they are feasible to implement and test.
Conversational guidance	Uses a conversational guide grounded in verified information and the student's profile.	Uses an interactive guidance sandbox alongside a dynamic decision summary.	Both support follow-up questions. The final implementation should restrict factual guidance to verified information.
Responsible guidance	Addresses uncertainty, unsupported claims, bias, privacy, and the limitations of recommendations.	Includes verification warnings, privacy protections, and safeguards against definitive judgments.	Both provide a useful foundation for responsible guidance, but their safeguards must be implemented and evaluated.
Feasibility within 14 weeks	Offers a rich feature set but includes many screens and interaction steps.	Offers fewer main stages but still requires integration of the frontend, backend, scoring logic, and conversational component.	Neither concept has been validated for the team's actual development capacity. A smaller MVP should be defined before implementation.

3. Selected Concept and Justification

3.1 Reasoning

Our proposed final direction is **Masari: An Explainable AI Guide for Exploring the Egyptian Baccalaureate Pathways**. It combines the detailed interaction and decision-support mechanisms proposed by Claude with the more compact structure and recommendation dashboard proposed by Gemini.

We selected this direction because the project needs to do more than suggest an academic pathway. It should help students understand their options, inspect the reasons behind suggestions, and remain in control of their decisions.

The final design will prioritize five core stages:

1. **Understand the System:** Present simplified explanations of the Egyptian Baccalaureate system using information verified against official sources. Unconfirmed details will be clearly flagged.
2. **Explore and Compare Pathways:** Allow students to explore available pathways and compare their documented characteristics using consistent criteria, without labeling one pathway as universally better than another.
3. **Assess Interests and Preferences:** Use a short, non-judgmental questionnaire to understand students' interests, preferred activities, and broad aspirations. Students may skip questions or revise their answers.
4. **Review Explainable Suggestions:** Present a small number of pathways worth exploring, with plain-language explanations of matching factors, mismatches, and uncertainties. The initial prototype will use documented, rule-based scoring rather than allowing an LLM to determine the ranking.
5. **Explore Alternatives and Reflect:** Let students ask follow-up questions, modify their answers, adjust the importance of selected factors, compare alternatives, and record their thoughts for further discussion.

Design Decisions

The final prototype will retain the most useful elements of both concepts: source verification, explainable recommendations, editable assessment inputs, and student-controlled exploration.

To keep the project feasible, the team will prioritize the core learning and recommendation journey before adding optional features such as PDF exports or more advanced scenario controls.

The prototype will not treat a match score as a prediction of a student's ability or future success. It will communicate uncertainty, avoid unsupported claims, and leave the final decision to the student.

All official pathway names, subject requirements, examination rules, and university admission information must be verified before being included as factual content. Proposed features and evaluation targets will be treated as plans until they are implemented and tested.

4. Prompt Log - Appendix A

4.1 The exact prompt

```
We are developing a university term project: an AI Guide for Grade 9 students transitioning to the Egyptian Baccalaureate system.
Project Idea:
Students may not understand how the new system works, what academic pathways are available, how they differ, or which options match their interests and strengths. Our AI Guide helps students understand the system, explore pathways, assess their interests, and make informed decisions.
The system follows five stages:
- Understand: Explain the system in simple, student-friendly language.
- Explore: Introduce available pathways and explain their differences.
- Assess: Use interactive questions to explore students' interests, strengths, and aspirations.
- Recommend: Suggest potentially suitable pathways and explain the reasons behind each recommendation.
- Guide: Let students ask follow-up questions, compare options, and reconsider recommendations before deciding.
Primary Human-AI Interaction Principle: Transparency, supported by Explainability and User Control and Agency.
Your Task
Act as an educational AI product designer. Create ONE complete concept draft for this AI Guide, describing how students interact with the system from understanding the Baccalaureate system to exploring pathways, completing an assessment, receiving recommendations, and discussing their options with the AI.
Structure your concept draft as follows:
1. Concept Name: Give the concept a short, memorable name.
2. Problem and Target Users: Describe the students' needs and the problem the system addresses.
3. Core Features: Explain the essential features supporting all five stages.
4. Core Interaction Flow: Present a numbered, step-by-step student journey. For each step, specify:
- Student action.
- What appears on the screen.
- System or AI response.
- What the student can do next.
5. Key Screens: Describe essential screens, interface elements, buttons, and navigation. Include simple text-based wireframes if useful.
6. Assessment and Recommendations: Explain how answers are analyzed, how pathways are matched to students' interests, and how recommendations are justified.
7. Transparency and User Control: Show how students inspect the reasons behind recommendations, compare alternatives, ask questions, and retain control over the final decision.
8. Responsible Guidance: Explain how the system handles uncertainty, avoids bias and unsupported conclusions, protects student information, and communicates its limitations.
9. MVP and Evaluation: Define a realistic prototype for a small student team, including essential components and measurable evaluation criteria.

Constraints
- Focus on the Egyptian Baccalaureate context and Grade 9 students.
- Do not invent official pathway names, requirements, examination rules, or government policies. Mark details requiring verification through authoritative Ministry of Education sources.
- Keep the design feasible for a university prototype.
- Make Transparency visible in actual screens and interactions.
- Explain recommendations clearly without presenting them as definitive judgments about students' abilities or futures.
- Let students compare options and make their own informed decisions.
- Avoid unnecessary features and excessive complexity.
- Use clear, professional English suitable for a university milestone report.
- Do not claim the system has been tested or validated without evidence.

Important: Output only the complete concept draft and its core interactive flow. Do not provide comparisons with other AI-generated concepts, a self-evaluation, strengths and weaknesses, or a recommendation about which concept should be selected..
```

Note: *The prompt above is quoted exactly as it was sent. The project's Human-AI Interaction principle was later fixed as User Control and Agency, with Transparency and Explainability supporting it.*

4.2 Prompt log records

#	Tool	Exact Prompt	What it gave	What we change and why
1.	**Claude** : **Sonnet 5.5**	The prompt documented in Section 4.1	A full concept draft, "Masar: The Transparent Pathway Guide", describing a detailed, flexible student journey with a reasoning panel, rule-based scoring, source verification flags, and a decision notebook. File: `Masari_Claude_Week_02.md`	Kept: the rule-based scoring, because documented rules let students inspect why a pathway is suggested, and the initial prototype will not let an LLM determine the ranking. Kept: the decision notebook, because it lets students record their thoughts and discuss their options with others before deciding.
2.	**Gemini Flash 3.6 Low**	The same prompt documented in Section 4.1	A full concept draft, "Masari: An Explainable AI Transition Guide for the Egyptian Baccalaureate", describing a five-stage flow with pathway comparison, an interactive assessment, a recommendation dashboard, and percentage-based affinity scores. File: `Masari_Gemini_Week_02.md`	Kept: the five-stage structure, because it is compact and easier to follow than a long journey. Kept: the recommendation dashboard, because it shows the factors behind each suggestion in one place. Not adopted: percentage-based affinity scores, because they imply more precision than the scoring method can justify (false precision).
3.	**Claude** : **Sonnet 5.5**	I am attaching our Week 2 document for the Masari project (HCAI Term Project 2026). Revise it by applying ONLY the changes below. Keep the existing structure, formatting, writing style, and level of detail. Do not remove relevant content and do not make unrelated changes. 1. Primary users: Parents and school counselors are now PRIMARY users, not "additional support". Update the Project Idea section (Target Users) and any other place that describes them as secondary. Users are: Grade 9 students, parents, and school counselors. 2. Principle consistency: The project's HAI principle is User Control and Agency. Where the	The revised Week 2 document with the seven requested changes applied, plus a short list of what was changed. File: `Week_02_revised.docx`	The document was revised as requested: primary users, principle note under Section 4.1, track and uniqueness in Section 5, completed prompt log, plain file names, adult-tester evaluation wording, and footer fix. The quoted prompt in Section 4.1 was left unchanged, and TA confirmation of uniqueness is marked [NEEDS TEAM INPUT].

		document or the quoted prompt mentions the principle, do not change the exact quoted prompt in Section 4.1 (it must stay as it was actually sent). Instead, add one short note under Section 4.1 saying that the principle was later fixed as User Control and Agency, with Transparency and Explainability supporting it. 3. Project Idea for Approval: Add (a) the track: "High schooler Guide", and (b) a short "Uniqueness" paragraph (3 to 4 sentences) explaining how our user group (Grade 9 students in Egypt exploring Baccalaureate pathways, with parents and school counselors as primary users) and idea (explainable, rule-based, student-controlled pathway exploration grounded in dated official sources) are distinct. Do not claim anything about other teams. Mark "Uniqueness confirmed with TA" as [NEEDS TEAM INPUT]. 4. Prompt Log (Section 4.2): - In the "What it gave" column, replace the bare file link with a one-sentence description of each output, followed by the file name Masari_Claude_Week_02.md / Masari_Gemini_Week_02.md. - Replace "N/A till now" in "What we changed and why" with what the team actually did, based only on Section 3 of this document: kept the rule-based scoring and decision notebook from Claude, kept the five-stage structure and recommendation dashboard from Gemini, and did not adopt percentage-based affinity scores because they imply false precision. State the reason for each. - Add a new entry (#3) for this revision request, quoting this prompt exactly as the "Exact Prompt", with a short description of what it gave and what changed. 5. File links: Replace the local C:\ paths with plain file names (Masari_Claude_Week_02.md and Masari_Gemini_Week_02.md) so they are not broken links. 6. Testing with minors: Change the Evaluation sentence to say that testing will be done with adult testers role-playing the user groups, and that no minors are involved. 7. Footer: Fix the footer so the page number format is correct (it currently shows "SEBA Page 1 of 1" on every page).		

		Rules: Do not invent decisions, evidence, or test results. Mark anything missing as [NEEDS TEAM INPUT]. Return a .docx file, then list in a short bullet list exactly what you changed.		

Note on the Concept Selection

The two outputs were used as alternative design proposals rather than implemented or validated systems. The selected direction combines useful elements from both proposals, subject to feasibility review and verification of official Baccalaureate information.

5. Project Idea for Approval

Masari — An Explainable AI Guide for Exploring the Egyptian Baccalaureate Pathways
Track: High schooler Guide
Target Users: Grade 9 students in Egypt preparing to enter the Egyptian Baccalaureate system, together with their parents and school counselors. All three groups are primary users of Masari.

Problem: Students may struggle to understand the available pathways, compare their differences, and relate their interests to their options. They need clear, reliable information to make informed decisions without relying on unsupported recommendations or external pressure.

Proposed Solution: Masari is an explainable AI guidance system that helps students explore and compare pathways through verified information and an interactive interest assessment. It provides rule-based suggestions with clear explanations, allowing students to modify preferences, compare alternatives, and ask follow-up questions. A conversational AI component may answer questions using verified information, while clearly communicating uncertainty and limitations.

Key Differentiator: Masari offers transparent, student-controlled recommendations rather than presenting a single pathway as the definitive choice. It supports informed exploration without predicting academic ability, university admission, or career success.

Uniqueness: Masari is built for a specific user group: Grade 9 students in Egypt who are exploring Baccalaureate pathways, together with their parents and school counselors as primary users. Its idea is explainable, rule-based, student-controlled pathway exploration, where every suggestion shows its reasons, students can edit their answers and the importance of factors, and factual content is grounded in dated official sources. Together, this user group and this idea define the scope of the project and what it sets out to offer.

MVP: The prototype will include a student-friendly system overview, pathway cards and comparisons, an interactive assessment, explainable rule-based suggestions, and a constrained conversational guide.

Evaluation: Usability tasks and structured feedback will assess recommendation explainability, user control, factual accuracy, and handling of uncertainty. These are planned evaluations; testing will be done with adult testers role-playing the user groups (Grade 9 students, parents, and school counselors), and no minors are involved.

Request for Approval: We request approval to develop Masari as a responsible, explainable AI guidance prototype. Official information will be verified against authoritative Ministry of Education sources, and implementation details will be finalized with the teaching assistant before development begins.